

Course Code : PHY-1201

Course Title : Physics II

Section : 2AF, CSE-54

Course Teacher : Dr. Muhammad Moazzam Hossen,
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Final
Math Problems for Practice
&
Syllabus

Book : Physics for Engineers Part II

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1.A) SOLID STATE PHYSICS

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Examples : 19.3, 19.4, 19 S, 19.6, 19.7, 19.8, 19.15, 19.16, 19.17, 19.18, 19.19

3.B) RELATIVITY

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Examples : 11.21, 11.22, 11.24, 11.25, 11.26, 11.28, 11.32, 11.34, 11.35

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Examples : 11.12, 11.13, 11.14, 11.15

4.B) MODERN PHYSICS.

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Examples : 12.4, 12.5 12.7

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Examples : 13.6, 13.7, 13.8, 13.10, 13.11

5.B) RADIOACTIVITY

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Examples : 16.1, 16.2, 16.3, 16.6, 16.7, 16.8, 16.10.

Syllabus :

PHY-1201

Section-B (SEE: 50 Marks)				
4	Structure of matter: Crystalline and non-crystalline solid, single crystal and polycrystalline solids, unit cell, Crystal systems, Coordination number, NaCl and CsCl structure, Packing fraction, Crystal planes and directions, Miller indices, calculation of inter-planar spacing for cubic system, Relation between inter-planar spacing & Miller indices.	04	CLO1 CLO2	
5	Defects & Bond in Solid: Different types of defects; Different types of bonds in solids, calculation of cohesive and bonding energy, Distinction between metal, insulator and semiconductor in terms of energy band.	06	CLO1 CLO2	
6	Relativity: <i>Postulates of special theory of relativity, Lorentz transformation, time dilation and length contraction, relativity of mass, energy-mass relation, energy-momentum relation.</i>	02	CLO1 CLO2	
7	Modern Physics: Bohr's atomic model, radius and energy of Hydrogen atom, atomic nucleus and binding energy, photo-electric effect, Compton Effect, De-Broglie waves, X-ray diffraction, atomic spectra and Zeeman Effect.	04	CLO1 CLO2	
8	Radioactivity: Definition, radioactive decay laws, half-life, mean life, alpha decay, beta decay, gamma decay, cross section, nuclear fission & fusion.	02	CLO1 CLO2	
		30		